

A Residue Reformulation of the Local Curve Identity

Conjecture 1. Let $l_1, l_2, l_3 \in \mathbb{Z}$ satisfy

$$l_1 + l_2 + l_3 = -2, \quad l_1 \geq l_2 \geq l_3,$$

and let $\lambda_1, \lambda_2, \lambda_3$ satisfy

$$\lambda_1 + \lambda_2 + \lambda_3 = 0.$$

Set

$$X_1 = \lambda_1^{-l_1-1} \lambda_2^{-l_2-1} \lambda_3^{-l_3-1}.$$

Define

$$\bar{l} = \begin{cases} l, & l \geq 0, \\ -l-1, & l < 0, \end{cases} \quad \sigma(m) := \sum_{i=1}^{m+1} (-1)^{i-1} (m - (i-1))^2 = \frac{m(m+1)}{2}.$$

Then

$$Y = \frac{1}{8} \lambda_1^{-2l_1-1} \lambda_2^{-2l_2-1} \lambda_3^{-2l_3-1} \left(\sigma(\bar{l}_1) \lambda_1^{-2} + \sigma(\bar{l}_2) \lambda_2^{-2} + \sigma(\bar{l}_3) \lambda_3^{-2} + l_1 l_2 \lambda_1^{-1} \lambda_2^{-1} + l_2 l_3 \lambda_2^{-1} \lambda_3^{-1} + l_1 l_3 \lambda_1^{-1} \lambda_3^{-1} \right).$$

Further, define

$$X_2 = -\lambda_1^{-2l_1-2} \lambda_2^{-2l_2-2} (\lambda_1 + \lambda_2)^{-2l_3-2} \left(\sum_{\substack{1 \leq k \leq l_1 \\ k \equiv l_1 \pmod{2}}} A(l_1, l_2, l_3, k) + \sum_{\substack{1 \leq k \leq l_2 \\ k \equiv l_2 \pmod{2}}} B(l_1, l_2, l_3, k) \right),$$

where

$$A(l_1, l_2, l_3, k) = \text{Res}_{h=0} \left[h^{-k} (h - \lambda_1)^2 (h + \lambda_2)^{k+l_2} (h + \lambda_1 - \lambda_2)^{k+l_3} \right. \\ \left. \times (h - \lambda_1 + \lambda_2)^{l_1-l_2-k} (h - 2\lambda_1 - \lambda_2)^{l_1-l_3-k} (h - 2\lambda_1)^{k-2-2l_1} \right],$$

and

$$B(l_1, l_2, l_3, k) = \text{Res}_{h=0} \left[h^{-k} (h - \lambda_2)^2 (h + \lambda_1)^{k+l_1} (h + \lambda_2 - \lambda_1)^{k+l_3} \right. \\ \left. \times (h - \lambda_2 + \lambda_1)^{l_2-l_1-k} (h - 2\lambda_2 - \lambda_1)^{l_2-l_3-k} (h - 2\lambda_2)^{k-2-2l_2} \right].$$

Prove the following identity:

$$Y = X_2 + \frac{1}{8} X_1.$$